

## Unit 36: Using Basic Integration Formulas

### 1. The idea, in plain words

You now have a whole toolkit of antiderivative formulas scattered across several units — the power rule, trig forms,  $e^x$ ,  $\ln|x|$ , and (from Unit 35) the inverse trig forms. This unit is about recognizing which formula fits a given integral, and — when the integral doesn't quite match any formula as written — reshaping it with algebra until it does.

The biggest new skill here: completing the square to unlock a hidden arcsin or arctan form. Whenever you see a quadratic expression sitting under a square root, or sitting alone in a denominator, and it doesn't factor nicely, try completing the square. This rewrites the quadratic as  $(x-h)^2$  plus or minus a constant — and that shape is exactly what the arcsin and arctan antiderivative formulas are built around.

Picture it like fitting a puzzle piece. The formulas  $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$  and  $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$  only fit a very specific shape: a perfect square subtracted from (or added to) a constant. A messy quadratic like  $x^2 - 6x + 13$  doesn't look like that shape at first glance — but after completing the square, it becomes  $(x-3)^2 + 4$ , which is exactly that shape (with  $u = x-3$  standing in for the variable).

### 2. Toolbox

The basic formulas, all in one place:

$$\begin{aligned} \int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) & \int \frac{1}{x} dx &= \ln|x| + C & \int e^x dx &= e^x + C \\ \int \sin x dx &= -\cos x + C & \int \cos x dx &= \sin x + C \\ \int \frac{dx}{\sqrt{a^2-x^2}} &= \arcsin\left(\frac{x}{a}\right) + C & \int \frac{dx}{a^2+x^2} &= \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C \end{aligned}$$

Completing the square, step by step (for a quadratic  $x^2 + bx + c$ ):

1. Take half of  $b$ , square it:  $\left(\frac{b}{2}\right)^2$ .
2. Add and subtract that amount:  $x^2 + bx + \left(\frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$ .
3. The first three terms form a perfect square:  $\left(x + \frac{b}{2}\right)^2$ .
4. Combine the leftover constants.

Matching the completed-square form to the right formula:

- If it ends up as (constant)  $-(x-h)^2$  under a square root  $\rightarrow$  use the arcsin formula, with  $u = x-h$ .
- If it ends up as  $(x-h)^2 +$  (constant) in a plain denominator  $\rightarrow$  use the arctan formula, with  $u = x-h$ .

Don't forget the substitution when the coefficient on  $x$  isn't 1 — if the quadratic has a coefficient like  $4x^2$  instead of plain  $x^2$ , a quick substitution (e.g.,  $u = 2x$ ) usually cleans it up before completing the square.

### 3. Common mistakes

- Forgetting the scaling factor “ $a$ ” in the arcsin/arctan formulas. These formulas aren't just  $\arcsin(x)$  or  $\arctan(x)$  unless  $a = 1$  — always divide by the correct  $a$  both inside the inverse trig function and (for arctan) out front.
- Sign errors while completing the square, especially when a leading negative sign is involved (like  $-x^2 + 4x - 3$ ) — factor the negative out carefully before completing the square on what's left.
- Misidentifying which formula applies. A quadratic under a square root pairs with arcsin; a quadratic alone in a denominator (no square root) pairs with arctan. Mixing these up leads to the wrong antiderivative shape entirely.
- Forgetting to substitute back at the end, if you introduced a  $u = x - h$  substitution partway through.
- Forgetting to double check your completed-square work by mentally re-expanding it — a small arithmetic slip here throws off everything that follows.

### 4. Problem Set

Warm-up

1.  $\int \frac{dx}{\sqrt{1-x^2}}$
2.  $\int \frac{dx}{1+x^2}$
3.  $\int x^4 dx$
4.  $\int e^{5x} dx$
5.  $\int \cos(4x) dx$
6.  $\int \frac{dx}{\sqrt{4-x^2}}$

Standard

7.  $\int \frac{dx}{9+x^2}$
8.  $\int \frac{dx}{\sqrt{16-x^2}}$
9.  $\int \frac{3dx}{x^2+4}$
10.  $\int \frac{dx}{\sqrt{9-4x^2}}$
11. Complete the square: rewrite  $x^2 - 6x + 13$  in the form  $(x-h)^2 + k$ .
12.  $\int \frac{dx}{x^2 - 6x + 13}$  (use your result from Problem 11)

13.  $\int \frac{dx}{\sqrt{-x^2 + 2x + 3}}$

Challenge

14.  $\int \frac{dx}{\sqrt{-x^2 + 4x - 3}}$

15.  $\int \frac{dx}{x^2 + 4x + 13}$

16.  $\int \frac{dx}{\sqrt{8 - 2x - x^2}}$

17.  $\int \frac{(2x + 3) dx}{x^2 + 4}$  (split into two pieces first!)

18.  $\int \frac{dx}{\sqrt{3 - 2x - x^2}}$

Applied

19. A particle's velocity is  $v(t) = \frac{1}{\sqrt{4 - t^2}}$ . Find the position function  $s(t)$ , given  $s(0) = 0$ .

20. An electrical current model is  $I(x) = \frac{1}{x^2 + 9}$ . Find the total charge  $\int_0^3 I(x) dx$ .

21. A structural deflection model requires evaluating  $\int \frac{dx}{\sqrt{-x^2 + 6x - 5}}$ . Complete the square and find the general antiderivative.

22. Evaluate  $\int_{-2}^2 \frac{dx}{\sqrt{4 - x^2}}$ .